Chapter 5, Section 5

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1. Let f be a rational function on the surface X. Show that it is possible to "resolve the singularities of f" in the following sense: there is a birational moprhism $g : X' \to X$ so that f induces a morphism of X' to $\mathbb{P}^1$.

Proof. Let $f : X \to \mathbb{P}^1$ be a rational map, and write the divisor of f as $(f) = \sum n_i C_i$. □

5. Let C be a curve, and let $\pi : X \to C$ and $\pi' : X' \to C$ be two geometrically ruled surfaces over C. Show that there is a finite sequence of elementary transformations (5.7.1) which transform X into X'.

Proof. □

6. Let X be a surface with function field K. Show that every valuation ring R of K/k is one of the three kinds described in (II, Ex 4.12).

Proof. □